

MARK SCHEME for the October/November 2008 question paper

9709 MATHEMATICS

9709/02

Paper 2, maximum raw mark 50

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began.

All Examiners are instructed that alternative correct answers and unexpected approaches in candidates' scripts must be given marks that fairly reflect the relevant knowledge and skills demonstrated.

Mark schemes must be read in conjunction with the question papers and the report on the examination.

- CIE will not enter into discussions or correspondence in connection with these mark schemes.

CIE is publishing the mark schemes for the October/November 2008 question papers for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level syllabuses and some Ordinary Level syllabuses.



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Mark Scheme Notes

Marks are of the following three types:

- M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

- A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

- B Mark for a correct result or statement independent of method marks.

- When a part of a question has two or more "method" steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly when there are several B marks allocated. The notation DM or DB (or dep*) is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

- The symbol $\checkmark$ implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only. A and B marks are not given for fortuitously "correct" answers or results obtained from incorrect working.

- Note: B2 or A2 means that the candidate can earn 2 or 0.
 B2/1/0 means that the candidate can earn anything from 0 to 2.

The marks indicated in the scheme may not be subdivided. If there is genuine doubt whether a candidate has earned a mark, allow the candidate the benefit of the doubt. Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored.

- Wrong or missing units in an answer should not lead to the loss of a mark unless the scheme specifically indicates otherwise.

- For a numerical answer, allow the A or B mark if a value is obtained which is correct to 3 s.f., or which would be correct to 3 s.f. if rounded (1 d.p. in the case of an angle). As stated above, an A or B mark is not given if a correct numerical answer arises fortuitously from incorrect working. For Mechanics questions, allow A or B marks for correct answers which arise from taking g equal to 9.8 or 9.81 instead of 10.

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The following abbreviations may be used in a mark scheme or used on the scripts:

AEF	Any Equivalent Form (of answer is equally acceptable)
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
BOD	Benefit of Doubt (allowed when the validity of a solution may not be absolutely clear)
CAO	Correct Answer Only (emphasising that no "follow through" from a previous error is allowed)
CWO	Correct Working Only - often written by a 'fortuitous' answer
ISW	Ignore Subsequent Working
MR	Misread
PA	Premature Approximation (resulting in basically correct work that is insufficiently accurate)
SOS	See Other Solution (the candidate makes a better attempt at the same question)
SR	Special Ruling (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)

Penalties

MR -1	A penalty of MR -1 is deducted from A or B marks when the data of a question or part question are genuinely misread and the object and difficulty of the question remain unaltered. In this case all A and B marks then become "follow through $\sqrt{}$ " marks. MR is not applied when the candidate misreads his own figures - this is regarded as an error in accuracy. An MR-2 penalty may be applied in particular cases if agreed at the coordination meeting.
PA -1	This is deducted from A or B marks in the case of premature approximation. The PA -1 penalty is usually discussed at the meeting.

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- 1 *EITHER:* State or imply non-modular inequality $(x - 3)^2 > (2x)^2$ or corresponding quadratic equation or pair of linear equations $(x - 3) = \pm 2x$ M1
 Make reasonable solution attempt at a 3-term quadratic, or solve two linear equations M1
 Obtain critical values $x = 1$ and $x = -3$ A1
 State answer $-3 < x < 1$ A1
- OR:* Obtain critical value $x = -3$ from a graphical method, or by inspection, or by solving a linear inequality or linear equation B1
 Obtain the critical value $x = 1$ similarly B2
 State answer $-3 < x < 1$ B1 [4]
- 2 (i) Substitute $x = -2$ and equate result to zero, or divide by $x + 2$ and equate constant remainder to zero M1
 Obtain answer $a = -13$ A1 [2]
- (ii) Obtain quadratic factor $2x^2 - 5x - 3$ B1
 Obtain linear factor $2x + 1$ B1
 Obtain linear factor $x - 3$ B1 [3]
 [Condone omission of repetition that $x + 2$ is a factor.]
 [If linear factors $2x + 1$, $x - 3$ obtained by remainder theorem or inspection, award B2 + B1.]
- 3 State or imply $\ln y = \ln A - x \ln b$ B1
 State $\ln A = 1.3$ B1
 Obtain $A = 3.67$ B1
 Form a numerical expression for the gradient of the line M1
 Obtain $b = 1.28$ A1 [5]
- 4 (i) Use correct $\sin(A + B)$ and $\cos(A + B)$ formulae M1
 Substitute exact values for $\sin 30^\circ$ etc. M1
 Obtain given answer correctly A1 [3]
- (ii) Solve for x M1
 Obtain answer $x = 10.9^\circ$ A1
 Obtain second answer $x = -169.1^\circ$ and no others in the range A1 [3]
 [Ignore answers outside the given range.]
- 5 Integrate and state term $\ln x$ B1
 Obtain term of the form $k \ln (2x + 1)$ M1
 State correct term $-2 \ln (2x + 1)$ A1
 Substitute limits correctly M1
 Use law for the logarithm of a product, quotient or power M1
 Obtain given answer correctly A1 [6]
- 6 At any stage, state the correct derivative of $e^{-\frac{1}{2}x}$ or $e^{\frac{1}{2}x}$ B1
 Use product or quotient rule M1
 Obtain correct first derivative in any form A1
 Obtain correct second derivative in any form B1 ✓
 Equate second derivative to zero and solve for x M1
 Obtain $x = 4$ A1
 Obtain $y = 4e^{-2}$, or equivalent A1 [7]

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- 7 (i) Make a recognizable sketch of a relevant graph, e.g. $y = \cos x$ or $y = 2 - 2x$ B1
Sketch a second relevant graph and justify the given statement B1 [2]
- (ii) Consider sign of $\cos x - (2 - 2x)$ at $x = 0.5$ and $x = 1$, or equivalent M1
Complete the argument correctly with appropriate calculations A1 [2]
- (iii) Show that the given equation is equivalent to $x = 1 - \frac{1}{2} \cos x$, or *vice versa* B1 [1]
- (iv) Use the iterative formula correctly at least once M1
Obtain final answer 0.58 A1
Show sufficient iterations to justify its accuracy to 2 d.p. or show there is a sign change in the interval (0.575, 0.585) B1 [3]
- 8 (i) (a) Use trig formulae and justify given result B1
(b) Use $1 - \sin^2 x = \cos^2 x$ M1
Obtain given result correctly A1 [3]
- (ii) Use quotient or chain rule M1
Obtain correct derivative in any form A1
Obtain given result correctly A1 [3]
- (iii) Obtain integral $\tan x + \sec x$ B1
Substitute limits correctly M1
Obtain exact answer $\sqrt{2}$, or equivalent A1 [3]